BLK II - UNIT 5: METOC Mission Products
c. Mission Execution Forecast (MEF)
<u>Introduction to the Mission Execution Forecast (MEF)</u>

The purpose of the Mission Execution Forecast (MEF) is to provide the mission planner with a detailed local area forecast as well as identifying any mission hazards in the local area and at the mission location. This forecast could range from a few hours to several hours depending on the requirements of your unit. MEFs at each location will vary in some aspect but each should, at a minimum, include forecasted weather for the local area, and a mission area forecast and hazards. This product is a more detailed forecast and is broken down hour by hour for local forecasts and in 6-hour blocks for the mission location to give the mission planner a quick overview of the expected conditions. Depending on the location of the mission area, your forecast could be different for the mission impact area.

<u>MEF Template & Formatting</u>
MEF Heading
Date: Your MEF should auto populate the day after the current day. The MEFs you create here in training will all start at 00Z which would cause the date to change to the next day on the calendar regardless of what the local time is. Always double check the date is correct just in case it does not auto populate.

Valid Time: The time is the time your MEF starts. The only time you will have to change the valid time is if you have to make an amendment to the MEF and the current time is after the start of the MEF.

Location: Put the station / ICAO of the forecast.

Forecaster: Put your initials in this block.

Mission Execution Forecast			
13 Dec 2025	VALID TIME: 0000Z	LOCATION: Keesler AFB / KBIX	FORECASTER: JC

Local Area Forecast
Time: All MEFs for this course will start at 00Z and forecast for 30 hours.

Wind: Report the wind data for each hour in encoded METAR format including any gust. Like the 5-Day, round the direction to the nearest 10 degrees and the speed to the nearest 5KT.

Sky Condition: Report the sky condition in encoded METAR format. Include any significant cloud types if forecasted.

Hazards: Report any additional hazards that could impact your station that is not in any other section of the MEF. This section will normally contain mission specific information for that hour.

Visibility: Report visibility in statute miles with 10 representing unrestricted visibility.

Weather: Any forecasted weather element or visibility restrictor using METAR format.

Temperature / Dewpoint: Report the temperature and dewpoint in Fahrenheit.

Altimeter Setting: Report the Altimeter setting in Inches of Mercury (INS). Include the decimal point.

Use the mission threshold stoplight chart for your specific mission to highlight the weather features that impact your mission in the appropriate color. This will provide a quick visual reference for what and when the impacts to the mission are.

LOCAL AREA FORECAST								
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	WX	TEMP (F)	DP (F)	ALSTG
0000Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	61	57	29.96
0100Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	60	57	29.95
0200Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	60	58	29.93
0300Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	59	58	29.93
0400Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	58	58	29.93
0500Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	56	55	29.93
0600Z	23010G15KT	OVC030	LGT ICG 120-250	6 SM	-RA	54	52	29.93
0700Z	23010G15KT	OVC025	LGT ICG 120-250	6 SM	-RA	52	51	29.93
0800Z	23010G15KT	OVC025	LGT ICG 120-250	6 SM	-RA	50	49	29.92
0900Z	23010G15KT	OVC025	MOD ICG 100-250	6 SM	-RA	48	47	29.92
1000Z	23010G15KT	OVC020	MOD ICG 100-250	6 SM	-RA	46	45	29.92
1100Z	23010G15KT	OVC020	MOD ICG 100-250	6 SM	-RA	44	43	29.90
1200Z	23010G20KT	OVC020	MOD ICG 100-250	6 SM	-RA	42	41	29.91
1300Z	23010G20KT	OVC020	MOD ICG 100-250	6 SM	-RA	40	38	29.91
1400Z	23010G20KT	OVC020	MOD ICG 100-250	5 SM	-RA	40	39	29.92
1500Z	23010G20KT	OVC020	MOD ICG 100-250	5 SM	-RA	40	39	29.92
1600Z	23010G20KT	OVC015	TSTMS WITHIN 25 SM / MOD ICG	4 SM	-RA	40	39	29.93
1700Z	23010G20KT	OVC015CB	TSTMS WITHIN 10 SM / MOD ICG	4 SM	VCTS RA	41	40	29.94
1800Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	41	29.94
1900Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.94
2000Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.94
2100Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.94
2200Z	24015G25KT	OVC008CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
2300Z	24015G25KT	OVC008CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
2400Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
0100Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.95
0200Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	41	41	29.96
0300Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	40	37	29.97
0400Z	25010G20KT	BKN030CB	TSTMS WITHIN 5 SM	3 SM	TSRA	40	35	29.99
0500Z	29010KT	BKN050	TSTMS WITHIN 25 SM	5 SM	-RA	38	31	30.00
0600Z	34010KT	BKN050	NONE	10 SM	NSW	35	25	30.01

Identifying Mission Impacts & Thresholds

Mission Type: You will include a forecast / outlook for each type of mission supported by your unit. Many units support more than one type of mission, so it is important to understand the types of missions your unit supports. This section is primarily used to inform mission planners of conditions that are outside the local area forecast. If your mission is in the local area, you do not need to include information in this section. Input the type of mission and location of the mission in this section.

Time: The time blocks for this section are broken up into 6-hour blocks.

Sky Condition / Hazards / Visibility / Weather: Use the same rules as the 5-day forecast and put the predominant conditions and/or the worst expected conditions for that time block. Color code the blocks using the same rules as the local area forecast.

MISSION WEATHER FORECAST					
MISSION TYPE	TIME	SKY CONDITION	HAZARDS	VIS	WX
FIXED WIND BARKSDALE AFB / KBAD	00-06Z	OVC020	MODERATE ICING 100-250	6 SM	-RA
	06-12Z	OVC010CB	THUNDERSTORMS WITHIN 5 SM	6 SM	TSTMS WITHIN 5 SM
	12-18Z	OVC010CB	THUNDERSTORMS WITHIN 5 SM	6 SM	TSTMS WITHIN 5 SM
	18-24Z	BKN050	NONE	10 SM	NONE

If you are forecasting for an oceanic or coastal mission, you will need to include the Sea State Data in the table on the MEF Template. Any other type of mission does not require information in that section.

SEA STATE DATA (If Applicable)	
WAVE HEIGHT (FT)	6ft
SEA SURFACE TEMP (F)	48F
SEA STATE / BEAUFORT #	5